



SECURITIES INDUSTRY AUTOMATION CORPORATION

National Market Systems

Common IP Multicast Distribution Network

Recipient Interface Specification

Date:
Version:

June 13, 2025
1.58

Revised by: SIAC Product Management

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REVISION LOG

Document Number: ml101830001

Title: National Market Systems Common IP Multicast Distribution Network
Recipient Interface Specification

Version	Date	Rev by	Pages affected	Comments
1.2	3/13/97	ML		Initial Release
1.3	11/24/97	ML	18	Typo, naming conformance issue
1.4	12/15/98	RL	All	Remove references to Bisync and make document present tense with respect to the NMS network; Remove appendix on required bandwidth
1.5	12/03/99	MC	1, 19	Added references for retransmission and playback data, and IP Group assignments
1.9	3/23/00	RL	All	Clean up and Reorganize document. Removed section on logical lines because it served no purpose Added more information on multicast protocols Remove references to Frame Relay support New IP source addresses for RAPs and MPR boxes added as an appendix
1.10	6/16/00	RL	Appendix C	Added new RAPS IPs for 2 new hosts: RAPSOPRA3 and RAPSOPRA4
1.11	10/5/00	RL	All (major)	Add time beacon specifications; add new CTS and CQS group numbers
1.12	10/18/00	RL	All (minor)	Incorporate review comments, fix page numbers
1.13	7/11/01	RL	Appendix C Appendix C.2	Added Appendix C.2. Added text to Appendix C.
1.14	11/15/01	RL	Entire document	Updates to reflect interface types available on a per service basis. T3 connectivity no longer available to new connections or upgrades.
1.15	12/04/01	RL	Appendices C and C.2	New OPRA addresses are in production and therefore deleted Appendix C. Changed name of Appendix C.2 to C.
1.16	08/28/02	RL, CE	All	Removed Legacy Options
1.17 1.18				Internal draft update, not distributed
1.19	12/17/02	RL, CE		Include BBO info
1.20	1/03/03	RL		Correct Typo in Appendix A

Version	Date	Rev by	Pages affected	Comments
1.21	3/06/03	RL, CE		Updated IP addresses for Multicast playback and retransmissions.
1.22	8/19/03	RL, CE		Removed non-BBO lines.
1.23	1/12/04	RL, CE		Transitioned to SFTI interconnection. Deleted Appendix B, renamed appendix C as B, and Appendix D as C.
1.24	1/20/05	LG	Appendix B; Throughout Document	Update of all source addresses. Removed reference to Site A and Site C. Replaced with Group A and Group B.
1.25	2/15/05	LG	Appendix B	Additional source addresses added; source addresses identified by A & B Streams.
1.26	11/02/05	MCP	Pages 1,3,4,6,7, 13 & 14 Appendix A & B	Page: 1, 4: Revised OPRA lines 1-9 to 'FCO 1' and OPRA 1-24 Page 3,4,6 & 7: Revised multicast totals Page 7, 13, &16: Added new address ranges Appendix A: Added new OPRA MCL addresses. Appendix B: Added two new source addresses.
1.27	10/10/06	MCP	Appendix B (Page 17)	Page 17 – Added new CTS/CQS source addresses - Deleted OPRA 8 Line network Addresses
1.28	12/13/06	MCP	Appendix B	Removed duplicate source IP's from CTS/CQS Production A & B stream as well as updated Time Beacon A & B stream
1.29	2/27/07	MCP	4.2 & Appendix A	Added OPRA FAST Multicast Addresses
1.30	1/23/08	LG	4.2 & Appendix A	Added OPRA FAST for Symbology Multicast Addresses (Reused ASCII)
1.31	11/24/08	LP	- Entire Document - Appendix A - Appendix B	-Eliminated FCO References -Renamed OPRA FAST Multicast Addresses to spare OPRA addresses -Added CTS/CQS Production, and Time Beacon Source IP Addresses (reassigned CTS, CQS, and OPRA Playback Source IPs)
1.32	2/23/09	LP	Appendix A, and Pgs: 8, 10, 12, 14,	Addition of new CTS/CQS Production, Retransmission and Playback Test Data Multicast Addresses/Ports
1.33	2/27/09	LP	Appendix A, and Pgs: 8, 9, 10, 12, 14,	-Revision of new CTS/CQS Production, Retransmission and Playback Test Data Multicast Addresses/Ports
1.34	2/18/10	WG	Table of Contents update	-Addition of new CTS/CQS/OPRA/Time Beacon Production Source IP Addresses

Version	Date	Rev by	Pages affected	Comments
			Appendix B (pages 29 – 38) Appendix C (pages 39 – 52) Appendix D (page 53)	-Addition of CTS/CQS/OPRA After Hours Playback Source IP Addresses -Addition of Production Retransmission and After Hours Playback Test Source IP Addresses - Relabeled as “Current” Source Addresses on the page header - Appendix C, relabeled as “Future” Source Addresses on page header - Created new Appendix D
1.35	6/1/10	MC	Appendix C : pages 39-40	- Corrected typos in Source addresses
1.36	8/19/10	MCP	Pages : Appendix ‘A’ (23 & 27) Pages : 8,10 & 11 IP Source Addresses	- Added OPRA 25-48 HSL’s - Revised OPRA “1-24” to 1-48” Appendix ‘B’ and Appendix C’ updated to Appendix ‘B’ = Primary Site and ‘C’ for Disaster Recovery IP Source Addresses
1.37	11/28/11	WG	Table of Contents Appendix B (pages 30 – 36) Appendix C (pages 39 – 41)	-Adjustment of page numbers -Addition of new Subnet “.53/.54” for CTS/CQS/OPRA/Time Beacon Production Source IP Addresses -Addition of new Subnet “.53/.54” for CTS/CQS/OPRA After Hours Playback and Retransmission Source IP Addresses
1.38	8/20/12	MCP	Table of Contents Page 15, 21, 24 Page 30	-Adjustment of page numbers -Addition of two Equity / Index Production test multicast lines (binary 3 and 4 to support binary output format over production) -Addition of a 48 line temporary dual network test group assignments (48-A Stream, 48-B stream) -Removed all references to SPARE addresses
1.39	6/18/13	MC	Page 28	-Addition of two Equity / Index Retransmission test multicast lines (binary 3 and 4 to support binary output format over production)
1.40	6/25/14	MCP	Page 9, 11, 12, 15, 21, 25 and 29	-addition of Extended Session
1.41	1/27/15	LP	Page 23, 27	-Removed Network A Index references (Indices Disseminated over Network B)
1.42	5/4/15	WG	Page 45 thru 49 Page 50 thru 54	- Added Appendix C - New Primary Data Center Source Addresses - Added Appendix C - New Disaster Recovery (Backup) Data Center Source Addresses

Version	Date	Rev by	Pages affected	Comments
			Page 35 Page 36	- Removed decommissioned source addresses on Network Bor subnets “.41 and .61”. - Removed decommissioned source addresses on Network Bor subnets “.42 and .62”.
1.43	2/21/17	WG	Pages 15, 21, 23, 24, 29 and 31 Pages 23 thru 28 & 30 thru 33 Pages 27 and 28 Pages 33 thru 46 Pages 49 thru 58	-Added dedicated binary parallel multicast addresses -Added to Multicast Addresses the reference page of Source Addresses -Added CQS and CTS Temporary Dual Network Test Group Assignments -Strike through applied to IP Addresses being decommissioned -Added to Appendix C (Source Addresses) the reference page for Multicast Addresses
1.43a	4/7/17	WG	Page 27 Page 27 and 28	-Corrected port assignments -Removed date from paragraph statement and added reference to updated notice
1.44	1/17/20	AA/MZ	All Pages Pages 19 thru 20	- Updated SFTI with ICE Global Network - Addition of Multicast Delivery over NMS Network
1.45	12/4/20	WG	Pages 16, 23 and 29 Pages 33 thru 47	-Added new Multicast IP Subnets -Added PILLAR SIP (CTS/CQS) Source and Multicast Addresses and Future OPRA Multicast Addresses
1.46	2/24/20	WG/AE	Appendix C	-Added redundant streams and revised addresses for Pillar SIP Retransmission and Playback test channels
1.47	8/6/20	AE	Whole document	-Removed all references to decommissioned Time Beacon; - Legacy CTA addresses removed
1.48	12/16/20	WG	Pages 16, 23 and 24 Page 35 Page 42	-Added new IP Ranges for OPRA Pillar -Added Source Addresses for OPRA Pillar -Added Appendix D with new Multicast IP Addresses for OPRA Pillar
1.49	1/22/21	WG	Page 45 Page 46	-Corrected Subnet IP Address - Added Missing Subnet IP Address
1.50	6/14/21	WG	Pages 44, 47 and 50 Pages 51 thru 53	-Removed inactive IP addresses from OPRA Extended Session -Added Appendix E for new OPRA GTH multicast IP Addresses
1.51	1/24/22	WG	Pages 10, 12, 16, 23, 24, 42 and 43	-Added references and tables for new Snapshot line and IP Addresses

Version	Date	Rev by	Pages affected	Comments
				-Removed old OPRA legacy IP Addresses and tables
1.52	3/8/23	WG	Pages 17, 24 and 25 Pages 43 thru 65	-Added New Address Ranges -Added New RP and Source Addresses -Added Appendix 1 thru 5: New OPRA, OPRA GTH, CTS and CQS multicast IP addresses
1.53	7/10/23	WG	Pages 26 thru 65	-Added decommission and activation dates -Corrected CTA Subnet from /24 to /25 -Added multicast line to Source Address table -Corrected port typo for OPRA Retransmission Line 41 and OPRA Playback Test line 42
1.54	9/11/23	WG	Pages 23 and 24	Table format change to list old and new IP addresses
1.55	2/5/24	WG	Pages 10, 12 and 13 Page 16, 26 thru 41 Pages 23 and 24 Pages 26 thru 43	Update Number of OPRA Lines and total number of redundancy Removal of old Multicast IPs for subnets; 224.0.86, .88, .89, .90, .92, .93, 94. 95 Removal of old CTA and OPRA addresses Move up new multicast IPs for subnets 224.0.200, .201, .202, .203, .204
1.56	5/1/24	WG	Page 26	Added OPRA GTH Line References to the Source Subnets table in the "Associated Multicast Lines" column
1.57	10/1/24	CS	Pages 10-16 (former pages 13-26)	-Revised document to reflect that the only direct multicast of NMS Services is via the NMS Network -Deleted as obsolete Section 1.2 "Multicast Primer"; Section 4.1 "IP Multicasting - Primer Part II"; Section 4.3 "UDP/IP Framing"; Section 4.4.4 "Multicast Delivery over ICE Global Network"; Section 4.5 "Logical Groups Mapping versus Physical Access Points", and Section 5 "Physical, Media Layer, and Network Connectivity" and renumbered remaining Sections
1.58	06/13/25	WG	Pages 37 and 41	CTS (p.37) and CQS (p.41) Playback Test addresses corrected

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1 Overview

This document provides the interface specifications for customers connecting to the National Market Systems (NMS) distribution network. This includes recipients of the Consolidated Tape System (CTS), Consolidated Quotation System (CQS), and the Options Price Reporting Authority (OPRA) real-time production data. Recipients should also use this specification for information related to receiving NMS real-time data retransmission and NMS after-hours playback data messages. The NMS distribution network disseminates all market data information in the form of multicast addressed IP datagrams.

In order to receive NMS Services (CT/CQ and OPRA), customers must connect to the NMS Network, which is dedicated for NMS Services only.

With respect to physical network connectivity, all data distributed by the NMS systems requires recipients to connect via the NMS Network. Recipients may choose to connect directly to the NMS Network or to connect indirectly using services from ICE Global Network (IGN) or a third-party service provider. For those choosing to connect via IGN, a separate interface specification, "[ICE Global Network & Colocation Technical Specifications](#)," should be referenced. For more information regarding the available connectivity options via IGN, please visit the ICE Global Network Website at <https://www.theice.com/data-services/global-network>, or contact ICE Global Network & Colocation Sales @ 1-770-661-0010 - Option 3 or send an email to iceglobalnetwork-info@theice.com.

1.1 Data Available via the NMS Network

NMS Real-Time Production Data

Two copies (Group A and Group B) of each NMS real-time production messages are available. These redundant copies are delivered via two distinct multicast data streams. For each unique NMS line (**CTS Network A lines 1-12, CTS Network B lines 1-12, CTS Index lines 1-2, CQS Network A lines 1-12, CQS Network B lines 1-12, OPRA lines 1-102 (Regular Session) and OPRA lines 201-204 (Global Trading Hours)**) there are two redundant multicast data streams. SIAC refers to these streams as the 'A' and 'B' streams. The 'A' stream is available from Group A and the 'B' stream from Group B. See Appendices A through E for the table of multicast group mappings. Therefore there will be 204 unique multicast groups allocated for the redundant delivery of these 156 lines ($156 * 2 = 312$).

NMS Real-Time Data Retransmission Data

The retransmission data streams are available via all network access points and are delivered via two distinct multicast data streams for **CTS Network A lines 1-12, CTS Network B lines 1-12, CTS Index lines 1-2, CQS Network A lines 1-12, CQS Network B lines 1-12, OPRA lines 1-102 (Regular Session) and OPRA lines 201-204 (Global Trading Hours)**. The recipient may choose to receive the retransmission data from either or both sites. See Appendices A through E for the table of multicast group mappings. Note: CTS Index lines 1-2 for Snapshot retransmissions is not supported.

NMS After-Hours Playback Data

Playback data is available in two methods:

- There is a set of Multicast data feeds dedicated for after-hours playback test data. This playback data is made available via two distinct multicast data streams for **CTS Network A lines 1-12, CTS Network B lines 1-12, CTS Index lines 1-2, CQS Network A lines 1-12, CQS Network B lines 1-12, OPRA lines 1-102 (Regular Session) and OPRA lines 201-204 (Global Trading Hours)**. See Appendices A through E for the table of multicast group mappings. Note: CTA After-Hours Playback Data for Snapshot is not supported.
- In addition to the playback test data groups, SIAC will continue to provide dual-sited redundant after-hours playback via the production system expressly for the purposes of redundancy testing.

Bandwidth Requirements

Bandwidth requirements change with time and recipients are encouraged to contact the ICE Global Network Help Desk and/or NMS Planning representatives regarding bandwidth requirements of each of the NMS services. For more information on the ICE Global Network and available services, please visit the ICE Global Network Website at <https://www.theice.com/data-services/global-network>, or contact ICE Global Network & Colocation Sales @ 1-770-661-0010 - Option 3 or send an email to iceglobalnetwork-info@theice.com.

Recipients should connect to the NMS Network via at least two access points in order to make full use of the redundant data feeds available for each service. Recipients not connecting directly should consult with their provider, whether IGN or a third party, regarding connectivity options.

Message Formats

For details of the message formats utilized by the CTS, CQS, and OPRA systems, please reference the following:

- CTS: CTS, Consolidated Tape System, Output Multicast Line, Interface Specification (www.ctaplan.com)
- CQS: CQS, Consolidated Quotation System, Output Multicast Line, Interface Specification (www.ctaplan.com)
- OPRA: OPRA, Options Price Reporting Authority, Data Recipient Interface Specification (www.opraplan.com)

Please note that the message format of retransmission and playback data is also governed by the documents listed above.

The remainder of this specification addresses the communications interfacing requirements for all data types.

Recipients connecting to the NMS Network via IGN are strongly recommended to consult the ICE Global Network and Colocation Technical Specifications, available here: https://www.ice.com/publicdocs/IGN_Colocation_US_Technical_Specifications.pdf, which provides additional information and considerations for receiving multicast services via ICE Global Network.

2 NMS Data Types

The NMS network distributes data via the multicast addressing and delivery protocols. Each of the three systems (CTS, CQS, and OPRA) has a unique set of multicast addresses assigned to each of its data “lines”. In each case there are redundant data streams provided for daytime production delivery of each line. The data lines for the three systems include CTS Network A lines 1-12, CTS Network B lines 1-12, CTS Index lines 1-2, CQS Network A lines 1-12, CQS Network B lines 1-12 and OPRA lines 1-102 (Regular Session) and OPRA lines 201-204 (Global Trading Hours). Therefore there will be 312 unique multicast groups allocated for the redundant delivery of these 156 lines ($156 * 2 = 312$).

- Each system line has two retransmission multicast address groups assigned to it. Entitled recipients have the option of subscribing to any retransmission line as needed.
- There are additional multicast group ID's allocated for supporting after-hours playback of the NMS data. Note that after-hours playback can also utilize the aforementioned production multicast groups.
- In all cases, recipients will only be permitted to receive data to which they are entitled.

Appendices A through E provides tables listing of all multicast group ID's. The tables also include a listing of the UDP destination port numbers assigned to each data stream. The NMS distribution system utilizes the UDP protocol at the IP transport layer. In order to provide the recipient community with the highest level of flexibility, the NMS systems have assigned a unique UDP destination port number to each multicast data stream. Note that the real-time redundant data streams use unique multicast addresses at the IP layer and unique UDP destination port numbers at the transport layer.

Recipient application software may make use of the UDP port mappings in order to multiplex between each of the data streams. Typically, applications use a “socket” programming interface which provides the means for requesting data on a per UDP port basis. If a port is not specified the application host's operating system might pass all IP broadcast data (including all multicast data) to a single process if the application has not specifically requested data on a per port basis. Please consult your application host's programming and system documentation for information particular to your environment.

3 Application Considerations

This section defines the application data framing and some of the key aspects of the IP distribution environment.

3.1 Application Encapsulation

In the IP environment the NMS application messages are also encapsulated in blocks, which in turn are encapsulated in an Ethernet frame as given in Figure 1 IP Data Block Format.

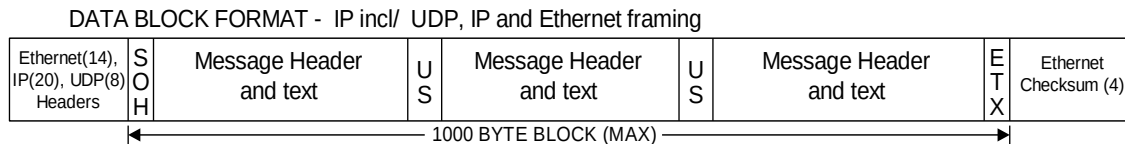


Figure 1 IP Data Block Format

There are several levels of encapsulation that occur within the Ethernet frame. The NMS data block, (which can be a maximum of 1000 bytes), is encapsulated within a UDP datagram, which in turn is encapsulated within an IP datagram which itself is encapsulated within an Ethernet frame. Each number shown in parentheses, e.g., IP (20) refers to the size of the particular header in bytes.

3.2 End to End Data Integrity

Integrity checking, on a per packet basis, is available via a checksum value in both the UDP header (**Error! Reference source not found.**) and the Ethernet frame check sequence.

In general, the Ethernet frame checksum validation is performed by the host's interface firmware and the IP checksum validation is performed within the TCP/IP stack and not by the application software.

Unlike TCP/IP based application services, the UDP/IP protocol has no "built-in" automatic retransmission functionality and therefore recipient host applications must examine the sequence numbers embedded within each NMS message on a per line basis in order to determine whether any data has been missed.

3.3 Line Concept

The term "line" refers to a specific logical data stream identified by the value pair formed by a unique IP multicast destination address and unique UDP destination port number.

Note that the following terms are all analogous to each other:

- multicast group
- multicast group ID
- multicast host group
- multicast host group ID
- multicast destination address

The NMS network currently utilizes 296 unique multicast group ID's for the purposes of providing NMS data to the recipient community. Each multicast group ID also has a UDP destination port number assigned to it, therefore each line of NMS data is uniquely identifiable by the value pair formed by its multicast group and UDP destination port number pair.

Appendices A through E provides the exact mappings of each line to its identifier pair. Reference section 1.2 Multicast Primer for a list summarizing the multicast groups.

4 Network Layer Connectivity

4.1 Multicast Addressing

The NMS network uses the following ranges, which are presented in further detail starting in Appendices A. Note that not all these addresses are in use.

SIAC Allocated Multicast IP Address Ranges

Starting Address	Ending Address
224.0.200.0	224.0.200.255
224.0.201.0	224.0.201.255
224.0.202.0	224.0.202.255
224.0.203.0	224.0.203.255
224.0.204.0	224.0.204.255
224.0.205.0	224.0.205.255
224.0.206.0	224.0.206.255
224.0.207.0	224.0.207.255

4.2 Multicast Address Use

The multicast group addresses used by SIAC for the dissemination of application data on this network have been registered with the Internet Assigned Numbering Authority (IANA). No recipient will be allowed to connect to the NMS distribution network if it is found that they are using any of these addresses for their own use.

For a list of these addresses please view <http://www.iana.org/assignments/multicast-addresses>.

4.2.1 IGMP

Internet Group Management Protocol (IGMP) is a protocol that end systems use to communicate with multicast compliant routers and is defined in RFC 1112. Recipient host systems that wish to subscribe to multicast groups must be fully compliant with this RFC.

4.2.2 Subscription Control

To receive the multicast packets, applications running on recipient end-stations issue IGMP subscription (or “join group”) packets on their locally attached LANs. The local router (which must also be multicast compliant) adds the multicast group to its registration table and begins to forward all packets destined to that group onto the LAN.

Recipients have the option of subscribing to any combination of multicast groups but as mentioned previously, SIAC will allow recipients to receive only those groups to which they have been entitled.

4.2.3 Multicast Delivery over NMS Network

NMS Services CT/CQ/OPRA are provided by NMS Network. NMS Network ports are classic Ethernet interface (un-tagged) configured for transporting aggregate multicast traffic for all services. NMS Network lowers the overall latency while maintaining the redundancy and fairness throughout the network.

To facilitate the delivery of Multicast data, NMS Network employs the use of multicast routing protocol. Protocol Independent Multicast (PIM) Sparse mode is utilized to accomplish this task. PIM Sparse mode is used to signal delivery and reception of multicast data. PIM Sparse mode requires the use of Rendezvous Point (RP) to act as meeting point between subscribers and listeners of multicast data. Router behind which interested listeners are, send PIM-Join signal to RP for interested multicast groups to join the Shared Tree for the reception of data. At this stage if the source(s) is already

publishing data and RP has established Multicast Tree towards the Source, the data starts flowing via RP Tree or Shared Tree.

NMS Network customers will have below two methods for receiving multicast data. This specification refers to the configuration of the customer router port directly connected to NMS Network. Customers can implement any network solution they wish beyond that interface. SIAC has no restrictions on the manner in which a customer designs its networks to support multicast reception. This applies to both protocol and physical topology perspectives. Customers are responsible for implementing a working design that best suits their environment.

Method I: Dynamic Multicast Routing - PIM Sparse Mode

- **PIM Sparse Mode:** Configure PIM Sparse Mode on the router that connects to Access Network.
- **Rendezvous Point:** RP-A and RP-B IP-Addresses are propagated via dynamic BGP unicast routing for A and B Multicast group mappings.
- **Shared Tree:** Upon receipt of PIM (*, G) Join, Shared Tree is established which is rooted at the Rendezvous Point for Any Source Multicast (ASM) delivery.
- **Shortest Path Tree:** Upon receipt of (S, G) Source / Group Join, Shortest Path Tree (SPT) is established for delivery of Multicast packets.

Method II: Static Multicast Routing - IGMP Static Groups

- SIAC, upon the request of the customer, will define IGMP static joins on the NMS Network Edge router connected to the customer. This will result in forwarding of multicast data for the multicast groups that are statically configured. Static IGMP joins are matched against entitled multicast groups to the customer edge router.
- Customer routers learn multicast source routes via dynamic unicast routing as described in section [4.1.1](#).

Customers can implement appropriate solutions they require on their edge router in order to correctly forward the multicast data into their networks. Typically, router vendors provide the option of forwarding the multicast data at the edge into their routing trees using the routing information learned via dynamic unicast routing.

4.2.3.1 Multicast Entitlement Control

Entitlement for Proprietary Market Data Feeds subscribers will be enforced at the NMS Network Edge Routers through the application of PIM Join Policies on the logical network interface of each individual Customer. The use of Policies provides the mechanism to control the transmission/reception of multicast services for dynamic subscribers. For customers that chose not to use dynamic multicast routing protocols, SIAC will define IGMP static groups for the entitled market data services and applied logical network interface of each individual Customer.

Ingress traffic filters on the Edge Router logical interfaces supporting multicast will silently discard any incoming packets except those used by the required multicast (PIM) and/or unicast routing protocols (BGP). These filters will also be used to protect each of the network components within the NMS Network from any customer-originated multicast traffic. Filters can be reconfigured dynamically to allow for timely re-provisioning of entitlements.

4.2.3.2 Multicast Data Retransmission

NMS Multicast data service provides an in-band retransmission request mechanism over a unicast TCP sockets based applications. NMS Multicast data service supports multicast retransmission over separate group ranges as defined in [Appendix-A](#) of this document. NMS Network subscribers are able to receive multicast retransmission over the same physical and/or logical interface as their primary production feeds.

Customers can implement any network solution they wish beyond the NMS Network interface. Beyond the NMS Network demarcation point, SIAC places no restrictions on the manner in which a customer designs its networks to support multicast reception. This is true from both from a protocol and physical topology perspectives. Customers are responsible for implementing a working design that best suits their environments.

4.3 Data Entitlement

For a recipient host system to receive a particular data stream it must subscribe to the data stream's corresponding multicast group ID via IGMP. Appendices A through C lists all multicast group ID assignments.

In order to restrict a recipient from subscribing to data streams that they are not entitled to, outbound packet filters are employed on SIAC's distribution routers interfaces connecting to the recipients. These filters block data from being sent to non-entitled recipients on a per service basis (CTS, CQS, and OPRA).

4.4 IP Addressing Considerations

Please consult the ICE Global Network interface specification for details.

4.5 Recipient Security

SIAC protects its network and hosts using several methods. Traffic filters and routing policies prevent sharing of information and data between entities connected to the NMS Network. Additional measures are in place as well, however these security measures maintain the integrity of SIAC's distribution environment by protecting SIAC's network and hosts from intentional or accidental access from within a recipient network.

These measures are in no way intended to provide the same level of security to the recipients themselves. If a recipient believes that additional security is required to protect their network they are encouraged to take action to implement additional security measures.

For the purposes of aiding in the implementation of security measures (e.g. traffic filters), the source IP addresses associated with the NMS systems have been provided in Appendices C and D.

5 Appendix A - NMS IP Multicast Addresses

This appendix contains the mapping of IP multicast group ID's (addresses) to the currently available data lines. To receive a particular data stream the recipient host system would typically subscribe to that particular multicast group ID. Two multicast group ID's are available for each real-time production data line. The data originating from Group A is generally referred to as the 'A' streams and the data from Group B as the 'B' streams. Also provided in the table are the UDP destination ports associated with each logical line.

The NMS data messages are encapsulated in an identical manner in both streams. For example, a datagram issued Group A on OPRA Line 2 destined to multicast group 224.0.204.2 will have a corresponding datagram (containing the identical UDP data payload, i.e. same NMS messages and same sequence number range) sourced from Group B destined to multicast group 224.0.206.2.

Multicast Address Ranges:

NMS PRODUCTION IP MULTICAST FEEDS GROUP A
224.0.200.0 - 224.0.200.15
224.0.200.96 - 224.0.200.99
224.0.202.0 - 224.0.202.31
224.0.202.32 - 224.0.202.63
224.0.203.0 - 224.0.203.31
224.0.203.32 - 224.0.203.63
224.0.204.0 - 224.0.204.127

NMS PRODUCTION IP MULTICAST FEEDS GROUP B
224.0.200.128 - 224.0.200.143
224.0.200.224 - 224.0.200.227
224.0.202.128 - 224.0.202.159
224.0.202.160 - 224.0.202.191
224.0.203.128 - 224.0.203.159
224.0.203.160 - 224.0.203.191
224.0.206.0 - 224.0.206.127

Appendix A - NMS IP Multicast Addresses, Cont'd

Retransmission and Playback Test Data, Single and Dual Sets

CTA supports dual data stream sets for the CTS and CQS retransmission and playback test data feeds (Appendix C). OPRA, supports dual data stream sets for retransmission and playback test data feeds (Appendix D). Playback data is only available after-hours.

Recipients wishing to receive retransmission and/or playback feeds must subscribe to the multicast feeds based on the addressing information shown in the following table.

Multicast Address Ranges:

NMS PRODUCTION IP MULTICAST FEEDS GROUP A
RETRANSMISSION GROUPS
224.0.200.100 - 224.0.200.103
224.0.200.16 - 224.0.200.31
224.0.201.0 - 224.0.201.31
224.0.201.32 - 224.0.201.63
224.0.202.64 - 224.0.202.95
224.0.202.96 - 224.0.202.127
224.0.203.64 - 224.0.203.95
224.0.203.96 - 224.0.203.127
224.0.204.128 - 224.0.204.255
PLAYBACK TEST GROUPS
224.0.200.104 - 224.0.200.107
224.0.200.32 - 224.0.200.47
224.0.200.56 - 224.0.200.57
224.0.200.58 - 224.0.200.59
224.0.200.60 - 224.0.200.61
224.0.200.62 - 224.0.200.63
224.0.205.0 - 224.0.205.127

NMS PRODUCTION IP MULTICAST FEEDS GROUP B
RETRANSMISSIONS GROUPS
224.0.200.228 - 224.0.200.231
224.0.200.144 - 224.0.200.159
224.0.201.128 - 224.0.201.159
224.0.201.160 - 224.0.201.191
224.0.202.192 - 224.0.202.223
224.0.202.224 - 224.-.202.255
224.0.203.192 - 224.0.203.223
224.0.203.224 - 224.0.203.255
224.0.206.128 - 224.0.206.255
PLAYBACK TEST GROUPS
224.0.200.160 - 224.0.200.175
224.0.200.184 - 224.0.200.185
224.0.200.186 - 224.0.200.187
224.0.200.188 - 224.0.200.189
224.0.200.190 - 224.0.200.191
224.0.200.232 - 224.0.200.235
224.0.207.0 - 224.0.207.127

6 Appendix 1 - Rendezvous Points and SIP Data Center Source Addresses

RENDEZVOUS POINTS FOR NMS NETWORK

NETWORK CONNECTION	RP Address - A STREAM	RP Address - B STREAM
NMS Network	159.125.57.222	159.125.57.223

MULTICAST DATA SOURCE SUBNETS FOR PRIMARY AND DISASTER DATA CENTERS

SYSTEM / DATA STREAM	PRIMARY SOURCE IP/SUBNET	DISASTER SOURCE IP/SUBNET	ASSOCIATED MULTICAST LINES
CTS & CQS Multicast A1	159.125.48.0/25	198.140.48.0/25	DATA STREAM A TAPE A LINES 1-12
CTS & CQS Multicast A2	159.125.48.128/25	198.140.48.128/25	DATA STREAM A TAPE B LINES 1-12 & INDEX LINES 1-2
CTS & CQS Multicast B1	159.125.49.0/25	198.140.49.0/25	DATA STREAM B TAPE A LINES 1-12
CTS & CQS Multicast B2	159.125.49.128/25	198.140.49.128/25	DATA STREAM B TAPE B LINES 1-12 & INDEX LINES 1-2
OPRA Multicast A1	162.69.44.0/24	162.68.44.0/24	DATA STREAM A OPRA LINES 1 - 24 OPRA GTH LINE 1
OPRA Multicast A2	162.69.45.0/24	162.68.45.0/24	DATA STREAM A OPRA LINES 25 - 48 OPRA GTH LINE 2
OPRA Multicast A3	162.69.46.0/24	162.68.46.0/24	DATA STREAM A OPRA LINES 49 - 72 OPRA GTH LINE 3
OPRA Multicast A4	162.69.47.0/24	162.68.47.0/24	DATA STREAM A OPRA LINES 73 - 96 & RESERVED LINES 220 - 225 OPRA GTH LINE 4
OPRA Multicast B1	162.69.52.0/24	162.68.52.0/24	DATA STREAM B OPRA LINES 1 - 24 OPRA GTH LINE 1
OPRA Multicast B2	162.69.53.0/24	162.68.53.0/24	DATA STREAM B OPRA LINES 25 - 48 OPRA GTH LINE 2
OPRA Multicast B3	162.69.54.0/24	162.68.54.0/24	DATA STREAM B OPRA LINES 49 - 72 OPRA GTH LINE 3
OPRA Multicast B4	162.69.55.0/24	162.68.55.0/24	DATA STREAM B OPRA LINES 73 - 96 & RESERVED LINES 220 - 225 OPRA GTH LINE 4

Note: Multicast Addresses can be referenced on the next following pages.

7 Appendix 2 - OPRA Regular Session Production, Real-Time IP Multicast Feeds, (Network 'A' only)

PRODUCT NAME:		OPRA Regular Session			
Network Subnet:		ONE (1) NETWORK SUBNET GROUP PER DATA STREAM			
PRODUCTION “DATA STREAM - A” SUBNET: 224.0.204.0/25					
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 1	224.0.204.1	45001	OPRA 49	224.0.204.49	45049
OPRA 2	224.0.204.2	45002	OPRA 50	224.0.204.50	45050
OPRA 3	224.0.204.3	45003	OPRA 51	224.0.204.51	45051
OPRA 4	224.0.204.4	45004	OPRA 52	224.0.204.52	45052
OPRA 5	224.0.204.5	45005	OPRA 53	224.0.204.53	45053
OPRA 6	224.0.204.6	45006	OPRA 54	224.0.204.54	45054
OPRA 7	224.0.204.7	45007	OPRA 55	224.0.204.55	45055
OPRA 8	224.0.204.8	45008	OPRA 56	224.0.204.56	45056
OPRA 9	224.0.204.9	45009	OPRA 57	224.0.204.57	45057
OPRA 10	224.0.204.10	45010	OPRA 58	224.0.204.58	45058
OPRA 11	224.0.204.11	45011	OPRA 59	224.0.204.59	45059
OPRA 12	224.0.204.12	45012	OPRA 60	224.0.204.60	45060
OPRA 13	224.0.204.13	45013	OPRA 61	224.0.204.61	45061
OPRA 14	224.0.204.14	45014	OPRA 62	224.0.204.62	45062
OPRA 15	224.0.204.15	45015	OPRA 63	224.0.204.63	45063
OPRA 16	224.0.204.16	45016	OPRA 64	224.0.204.64	45064
OPRA 17	224.0.204.17	45017	OPRA 65	224.0.204.65	45065
OPRA 18	224.0.204.18	45018	OPRA 66	224.0.204.66	45066
OPRA 19	224.0.204.19	45019	OPRA 67	224.0.204.67	45067
OPRA 20	224.0.204.20	45020	OPRA 68	224.0.204.68	45068
OPRA 21	224.0.204.21	45021	OPRA 69	224.0.204.69	45069
OPRA 22	224.0.204.22	45022	OPRA 70	224.0.204.70	45070
OPRA 23	224.0.204.23	45023	OPRA 71	224.0.204.71	45071
OPRA 24	224.0.204.24	45024	OPRA 72	224.0.204.72	45072
OPRA 25	224.0.204.25	45025	OPRA 73	224.0.204.73	45073
OPRA 26	224.0.204.26	45026	OPRA 74	224.0.204.74	45074
OPRA 27	224.0.204.27	45027	OPRA 75	224.0.204.75	45075
OPRA 28	224.0.204.28	45028	OPRA 76	224.0.204.76	45076
OPRA 29	224.0.204.29	45029	OPRA 77	224.0.204.77	45077
OPRA 30	224.0.204.30	45030	OPRA 78	224.0.204.78	45078
OPRA 31	224.0.204.31	45031	OPRA 79	224.0.204.79	45079
OPRA 32	224.0.204.32	45032	OPRA 80	224.0.204.80	45080
OPRA 33	224.0.204.33	45033	OPRA 81	224.0.204.81	45081
OPRA 34	224.0.204.34	45034	OPRA 82	224.0.204.82	45082
OPRA 35	224.0.204.35	45035	OPRA 83	224.0.204.83	45083
OPRA 36	224.0.204.36	45036	OPRA 84	224.0.204.84	45084
OPRA 37	224.0.204.37	45037	OPRA 85	224.0.204.85	45085
OPRA 38	224.0.204.38	45038	OPRA 86	224.0.204.86	45086
OPRA 39	224.0.204.39	45039	OPRA 87	224.0.204.87	45087
OPRA 40	224.0.204.40	45040	OPRA 88	224.0.204.88	45088
OPRA 41	224.0.204.41	45041	OPRA 89	224.0.204.89	45089
OPRA 42	224.0.204.42	45042	OPRA 90	224.0.204.90	45090
OPRA 43	224.0.204.43	45043	OPRA 91	224.0.204.91	45091
OPRA 44	224.0.204.44	45044	OPRA 92	224.0.204.92	45092
OPRA 45	224.0.204.45	45045	OPRA 93	224.0.204.93	45093
OPRA 46	224.0.204.46	45046	OPRA 94	224.0.204.94	45094
OPRA 47	224.0.204.47	45047	OPRA 95	224.0.204.95	45095
OPRA 48	224.0.204.48	45048	OPRA 96	224.0.204.96	45096

Appendix 2 - OPRA Regular Session Production, Real-Time IP Multicast Feeds, (Network 'A' only) - **CONTINUED**

PRODUCT NAME:	OPRA Regular Session	
	RESERVED FOR FUTURE USE	
PRODUCTION “DATA STREAM - A” SUBNET: 224.0.204.0/25		
Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 220	224.0.204.105	45105
OPRA 221	224.0.204.106	45106
OPRA 222	224.0.204.107	45107
OPRA 223	224.0.204.108	45108
OPRA 224	224.0.204.109	45109
OPRA 225	224.0.204.110	45110

Appendix 2 - OPRA Regular Session Production, Real-Time IP Multicast Feeds, (Network 'B' only)

PRODUCT NAME:		OPRA Regular Session			
Network Subnet:		ONE (1) NETWORK SUBNET GROUP PER DATA STREAM			
PRODUCTION “DATA STREAM - B” SUBNET: 224.0.206.0/25					
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 1	224.0.206.1	45001	OPRA 49	224.0.206.49	45049
OPRA 2	224.0.206.2	45002	OPRA 50	224.0.206.50	45050
OPRA 3	224.0.206.3	45003	OPRA 51	224.0.206.51	45051
OPRA 4	224.0.206.4	45004	OPRA 52	224.0.206.52	45052
OPRA 5	224.0.206.5	45005	OPRA 53	224.0.206.53	45053
OPRA 6	224.0.206.6	45006	OPRA 54	224.0.206.54	45054
OPRA 7	224.0.206.7	45007	OPRA 55	224.0.206.55	45055
OPRA 8	224.0.206.8	45008	OPRA 56	224.0.206.56	45056
OPRA 9	224.0.206.9	45009	OPRA 57	224.0.206.57	45057
OPRA 10	224.0.206.10	45010	OPRA 58	224.0.206.58	45058
OPRA 11	224.0.206.11	45011	OPRA 59	224.0.206.59	45059
OPRA 12	224.0.206.12	45012	OPRA 60	224.0.206.60	45060
OPRA 13	224.0.206.13	45013	OPRA 61	224.0.206.61	45061
OPRA 14	224.0.206.14	45014	OPRA 62	224.0.206.62	45062
OPRA 15	224.0.206.15	45015	OPRA 63	224.0.206.63	45063
OPRA 16	224.0.206.16	45016	OPRA 64	224.0.206.64	45064
OPRA 17	224.0.206.17	45017	OPRA 65	224.0.206.65	45065
OPRA 18	224.0.206.18	45018	OPRA 66	224.0.206.66	45066
OPRA 19	224.0.206.19	45019	OPRA 67	224.0.206.67	45067
OPRA 20	224.0.206.20	45020	OPRA 68	224.0.206.68	45068
OPRA 21	224.0.206.21	45021	OPRA 69	224.0.206.69	45069
OPRA 22	224.0.206.22	45022	OPRA 70	224.0.206.70	45070
OPRA 23	224.0.206.23	45023	OPRA 71	224.0.206.71	45071
OPRA 24	224.0.206.24	45024	OPRA 72	224.0.206.72	45072
OPRA 25	224.0.206.25	45025	OPRA 73	224.0.206.73	45073
OPRA 26	224.0.206.26	45026	OPRA 74	224.0.206.74	45074
OPRA 27	224.0.206.27	45027	OPRA 75	224.0.206.75	45075
OPRA 28	224.0.206.28	45028	OPRA 76	224.0.206.76	45076
OPRA 29	224.0.206.29	45029	OPRA 77	224.0.206.77	45077
OPRA 30	224.0.206.30	45030	OPRA 78	224.0.206.78	45078
OPRA 31	224.0.206.31	45031	OPRA 79	224.0.206.79	45079
OPRA 32	224.0.206.32	45032	OPRA 80	224.0.206.80	45080
OPRA 33	224.0.206.33	45033	OPRA 81	224.0.206.81	45081
OPRA 34	224.0.206.34	45034	OPRA 82	224.0.206.82	45082
OPRA 35	224.0.206.35	45035	OPRA 83	224.0.206.83	45083
OPRA 36	224.0.206.36	45036	OPRA 84	224.0.206.84	45084
OPRA 37	224.0.206.37	45037	OPRA 85	224.0.206.85	45085
OPRA 38	224.0.206.38	45038	OPRA 86	224.0.206.86	45086
OPRA 39	224.0.206.39	45039	OPRA 87	224.0.206.87	45087
OPRA 40	224.0.206.40	45040	OPRA 88	224.0.206.88	45088
OPRA 41	224.0.206.41	45041	OPRA 89	224.0.206.89	45089
OPRA 42	224.0.206.42	45042	OPRA 90	224.0.206.90	45090
OPRA 43	224.0.206.43	45043	OPRA 91	224.0.206.91	45091
OPRA 44	224.0.206.44	45044	OPRA 92	224.0.206.92	45092
OPRA 45	224.0.206.45	45045	OPRA 93	224.0.206.93	45093
OPRA 46	224.0.206.46	45046	OPRA 94	224.0.206.94	45094
OPRA 47	224.0.206.47	45047	OPRA 95	224.0.206.95	45095
OPRA 48	224.0.206.48	45048	OPRA 96	224.0.206.96	45096

Appendix 2 - OPRA Regular Session Production, Real-Time IP Multicast Feeds, (Network 'B' only) - **CONTINUED**

PRODUCT NAME:	OPRA Regular Session	
	RESERVED FOR FUTURE USE	
PRODUCTION “DATA STREAM - B” SUBNET: 224.0.206.0/25		
Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 220	224.0.206.105	45105
OPRA 221	224.0.206.106	45106
OPRA 222	224.0.206.107	45107
OPRA 223	224.0.206.108	45108
OPRA 224	224.0.206.109	45109
OPRA 225	224.0.206.110	45110

Appendix 2 - OPRA Regular Session Retransmission IP Multicast Feeds, (Network 'A' only)

PRODUCT NAME:		OPRA Regular Session			
Network Subnet:		ONE (1) NETWORK SUBNET GROUP PER DATA STREAM			
RETRANSMISSION “DATA STREAM - A” SUBNET: 224.0.204.128/25					
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 1	224.0.204.128	46001	OPRA 49	224.0.204.176	46049
OPRA 2	224.0.204.129	46002	OPRA 50	224.0.204.177	46050
OPRA 3	224.0.204.130	46003	OPRA 51	224.0.204.178	46051
OPRA 4	224.0.204.131	46004	OPRA 52	224.0.204.179	46052
OPRA 5	224.0.204.132	46005	OPRA 53	224.0.204.180	46053
OPRA 6	224.0.204.133	46006	OPRA 54	224.0.204.181	46054
OPRA 7	224.0.204.134	46007	OPRA 55	224.0.204.182	46055
OPRA 8	224.0.204.135	46008	OPRA 56	224.0.204.183	46056
OPRA 9	224.0.204.136	46009	OPRA 57	224.0.204.184	46057
OPRA 10	224.0.204.137	46010	OPRA 58	224.0.204.185	46058
OPRA 11	224.0.204.138	46011	OPRA 59	224.0.204.186	46059
OPRA 12	224.0.204.139	46012	OPRA 60	224.0.204.187	46060
OPRA 13	224.0.204.140	46013	OPRA 61	224.0.204.188	46061
OPRA 14	224.0.204.141	46014	OPRA 62	224.0.204.189	46062
OPRA 15	224.0.204.142	46015	OPRA 63	224.0.204.190	46063
OPRA 16	224.0.204.143	46016	OPRA 64	224.0.204.191	46064
OPRA 17	224.0.204.144	46017	OPRA 65	224.0.204.192	46065
OPRA 18	224.0.204.145	46018	OPRA 66	224.0.204.193	46066
OPRA 19	224.0.204.146	46019	OPRA 67	224.0.204.194	46067
OPRA 20	224.0.204.147	46020	OPRA 68	224.0.204.195	46068
OPRA 21	224.0.204.148	46021	OPRA 69	224.0.204.196	46069
OPRA 22	224.0.204.149	46022	OPRA 70	224.0.204.197	46070
OPRA 23	224.0.204.150	46023	OPRA 71	224.0.204.198	46071
OPRA 24	224.0.204.151	46024	OPRA 72	224.0.204.199	46072
OPRA 25	224.0.204.152	46025	OPRA 73	224.0.204.200	46073
OPRA 26	224.0.204.153	46026	OPRA 74	224.0.204.201	46074
OPRA 27	224.0.204.154	46027	OPRA 75	224.0.204.202	46075
OPRA 28	224.0.204.155	46028	OPRA 76	224.0.204.203	46076
OPRA 29	224.0.204.156	46029	OPRA 77	224.0.204.204	46077
OPRA 30	224.0.204.157	46030	OPRA 78	224.0.204.205	46078
OPRA 31	224.0.204.158	46031	OPRA 79	224.0.204.206	46079
OPRA 32	224.0.204.159	46032	OPRA 80	224.0.204.207	46080
OPRA 33	224.0.204.160	46033	OPRA 81	224.0.204.208	46081
OPRA 34	224.0.204.161	46034	OPRA 82	224.0.204.209	46082
OPRA 35	224.0.204.162	46035	OPRA 83	224.0.204.210	46083
OPRA 36	224.0.204.163	46036	OPRA 84	224.0.204.211	46084
OPRA 37	224.0.204.164	46037	OPRA 85	224.0.204.212	46085
OPRA 38	224.0.204.165	46038	OPRA 86	224.0.204.213	46086
OPRA 39	224.0.204.166	46039	OPRA 87	224.0.204.214	46087
OPRA 40	224.0.204.167	46040	OPRA 88	224.0.204.215	46088
OPRA 41	224.0.204.168	46041	OPRA 89	224.0.204.216	46089
OPRA 42	224.0.204.169	46042	OPRA 90	224.0.204.217	46090
OPRA 43	224.0.204.170	46043	OPRA 91	224.0.204.218	46091
OPRA 44	224.0.204.171	46044	OPRA 92	224.0.204.219	46092
OPRA 45	224.0.204.172	46045	OPRA 93	224.0.204.220	46093
OPRA 46	224.0.204.173	46046	OPRA 94	224.0.204.221	46094
OPRA 47	224.0.204.174	46047	OPRA 95	224.0.204.222	46095
OPRA 48	224.0.204.175	46048	OPRA 96	224.0.204.223	46096

Appendix 2 - OPRA Regular Session Retransmission IP Multicast Feeds, (Network 'A' only) - **CONTINUED**

PRODUCT NAME:	OPRA Regular Session		
	RESERVED FOR FUTURE USE		
RETRANSMISSION “DATA STREAM - A” SUBNET: 224.0.204.128/25			
Originated Data Line	Multicast Group Address		Multicast Group Port
OPRA 220	224.0.204.232		46232
OPRA 221	224.0.204.233		46233
OPRA 222	224.0.204.234		46234
OPRA 223	224.0.204.235		46235
OPRA 224	224.0.204.236		46236
OPRA 225	224.0.204.237		46237

Appendix 2 - OPRA Regular Session Retransmission IP Multicast Feeds, (Network 'B' only)

PRODUCT NAME:		OPRA Regular Session			
Network Subnet:		ONE (1) NETWORK SUBNET GROUP PER DATA STREAM			
RETRANSMISSION “DATA STREAM - B” SUBNET: 224.0.206.128/25					
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 1	224.0.206.128	46001	OPRA 49	224.0.206.176	46049
OPRA 2	224.0.206.129	46002	OPRA 50	224.0.206.177	46050
OPRA 3	224.0.206.130	46003	OPRA 51	224.0.206.178	46051
OPRA 4	224.0.206.131	46004	OPRA 52	224.0.206.179	46052
OPRA 5	224.0.206.132	46005	OPRA 53	224.0.206.180	46053
OPRA 6	224.0.206.133	46006	OPRA 54	224.0.206.181	46054
OPRA 7	224.0.206.134	46007	OPRA 55	224.0.206.182	46055
OPRA 8	224.0.206.135	46008	OPRA 56	224.0.206.183	46056
OPRA 9	224.0.206.136	46009	OPRA 57	224.0.206.184	46057
OPRA 10	224.0.206.137	46010	OPRA 58	224.0.206.185	46058
OPRA 11	224.0.206.138	46011	OPRA 59	224.0.206.186	46059
OPRA 12	224.0.206.139	46012	OPRA 60	224.0.206.187	46060
OPRA 13	224.0.206.140	46013	OPRA 61	224.0.206.188	46061
OPRA 14	224.0.206.141	46014	OPRA 62	224.0.206.189	46062
OPRA 15	224.0.206.142	46015	OPRA 63	224.0.206.190	46063
OPRA 16	224.0.206.143	46016	OPRA 64	224.0.206.191	46064
OPRA 17	224.0.206.144	46017	OPRA 65	224.0.206.192	46065
OPRA 18	224.0.206.145	46018	OPRA 66	224.0.206.193	46066
OPRA 19	224.0.206.146	46019	OPRA 67	224.0.206.194	46067
OPRA 20	224.0.206.147	46020	OPRA 68	224.0.206.195	46068
OPRA 21	224.0.206.148	46021	OPRA 69	224.0.206.196	46069
OPRA 22	224.0.206.149	46022	OPRA 70	224.0.206.197	46070
OPRA 23	224.0.206.150	46023	OPRA 71	224.0.206.198	46071
OPRA 24	224.0.206.151	46024	OPRA 72	224.0.206.199	46072
OPRA 25	224.0.206.152	46025	OPRA 73	224.0.206.200	46073
OPRA 26	224.0.206.153	46026	OPRA 74	224.0.206.201	46074
OPRA 27	224.0.206.154	46027	OPRA 75	224.0.206.202	46075
OPRA 28	224.0.206.155	46028	OPRA 76	224.0.206.203	46076
OPRA 29	224.0.206.156	46029	OPRA 77	224.0.206.204	46077
OPRA 30	224.0.206.157	46030	OPRA 78	224.0.206.205	46078
OPRA 31	224.0.206.158	46031	OPRA 79	224.0.206.206	46079
OPRA 32	224.0.206.159	46032	OPRA 80	224.0.206.207	46080
OPRA 33	224.0.206.160	46033	OPRA 81	224.0.206.208	46081
OPRA 34	224.0.206.161	46034	OPRA 82	224.0.206.209	46082
OPRA 35	224.0.206.162	46035	OPRA 83	224.0.206.210	46083
OPRA 36	224.0.206.163	46036	OPRA 84	224.0.206.211	46084
OPRA 37	224.0.206.164	46037	OPRA 85	224.0.206.212	46085
OPRA 38	224.0.206.165	46038	OPRA 86	224.0.206.213	46086
OPRA 39	224.0.206.166	46039	OPRA 87	224.0.206.214	46087
OPRA 40	224.0.206.167	46040	OPRA 88	224.0.206.215	46088
OPRA 41	224.0.206.168	46041	OPRA 89	224.0.206.216	46089
OPRA 42	224.0.206.169	46042	OPRA 90	224.0.206.217	46090
OPRA 43	224.0.206.170	46043	OPRA 91	224.0.206.218	46091
OPRA 44	224.0.206.171	46044	OPRA 92	224.0.206.219	46092
OPRA 45	224.0.206.172	46045	OPRA 93	224.0.206.220	46093
OPRA 46	224.0.206.173	46046	OPRA 94	224.0.206.221	46094
OPRA 47	224.0.206.174	46047	OPRA 95	224.0.206.222	46095
OPRA 48	224.0.206.175	46048	OPRA 96	224.0.206.223	46096

Appendix 2 - OPRA Regular Session Retransmission IP Multicast Feeds, (Network 'B' only) - **CONTINUED**

PRODUCT NAME:	OPRA Regular Session	
	RESERVED FOR FUTURE USE	
RETRANSMISSION “DATA STREAM - B” SUBNET: 224.0.206.128/25		
Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 220	224.0.206.232	46232
OPRA 221	224.0.206.233	46233
OPRA 222	224.0.206.234	46234
OPRA 223	224.0.206.235	46235
OPRA 224	224.0.206.236	46236
OPRA 225	224.0.206.237	46237

Appendix 2 - OPRA Regular Session Playback Test IP Multicast Feeds, (Network 'A' only)

PRODUCT NAME:		OPRA Regular Session			
Network Subnet:		ONE (1) NETWORK SUBNET GROUP PER DATA STREAM			
PLAYBACK TEST “DATA STREAM - A” SUBNET: 224.0.205.0/25					
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 1	224.0.205.1	47001	OPRA 49	224.0.205.49	47049
OPRA 2	224.0.205.2	47002	OPRA 50	224.0.205.50	47050
OPRA 3	224.0.205.3	47003	OPRA 51	224.0.205.51	47051
OPRA 4	224.0.205.4	47004	OPRA 52	224.0.205.52	47052
OPRA 5	224.0.205.5	47005	OPRA 53	224.0.205.53	47053
OPRA 6	224.0.205.6	47006	OPRA 54	224.0.205.54	47054
OPRA 7	224.0.205.7	47007	OPRA 55	224.0.205.55	47055
OPRA 8	224.0.205.8	47008	OPRA 56	224.0.205.56	47056
OPRA 9	224.0.205.9	47009	OPRA 57	224.0.205.57	47057
OPRA 10	224.0.205.10	47010	OPRA 58	224.0.205.58	47058
OPRA 11	224.0.205.11	47011	OPRA 59	224.0.205.59	47059
OPRA 12	224.0.205.12	47012	OPRA 60	224.0.205.60	47060
OPRA 13	224.0.205.13	47013	OPRA 61	224.0.205.61	47061
OPRA 14	224.0.205.14	47014	OPRA 62	224.0.205.62	47062
OPRA 15	224.0.205.15	47015	OPRA 63	224.0.205.63	47063
OPRA 16	224.0.205.16	47016	OPRA 64	224.0.205.64	47064
OPRA 17	224.0.205.17	47017	OPRA 65	224.0.205.65	47065
OPRA 18	224.0.205.18	47018	OPRA 66	224.0.205.66	47066
OPRA 19	224.0.205.19	47019	OPRA 67	224.0.205.67	47067
OPRA 20	224.0.205.20	47020	OPRA 68	224.0.205.68	47068
OPRA 21	224.0.205.21	47021	OPRA 69	224.0.205.69	47069
OPRA 22	224.0.205.22	47022	OPRA 70	224.0.205.70	47070
OPRA 23	224.0.205.23	47023	OPRA 71	224.0.205.71	47071
OPRA 24	224.0.205.24	47024	OPRA 72	224.0.205.72	47072
OPRA 25	224.0.205.25	47025	OPRA 73	224.0.205.73	47073
OPRA 26	224.0.205.26	47026	OPRA 74	224.0.205.74	47074
OPRA 27	224.0.205.27	47027	OPRA 75	224.0.205.75	47075
OPRA 28	224.0.205.28	47028	OPRA 76	224.0.205.76	47076
OPRA 29	224.0.205.29	47029	OPRA 77	224.0.205.77	47077
OPRA 30	224.0.205.30	47030	OPRA 78	224.0.205.78	47078
OPRA 31	224.0.205.31	47031	OPRA 79	224.0.205.79	47079
OPRA 32	224.0.205.32	47032	OPRA 80	224.0.205.80	47080
OPRA 33	224.0.205.33	47033	OPRA 81	224.0.205.81	47081
OPRA 34	224.0.205.34	47034	OPRA 82	224.0.205.82	47082
OPRA 35	224.0.205.35	47035	OPRA 83	224.0.205.83	47083
OPRA 36	224.0.205.36	47036	OPRA 84	224.0.205.84	47084
OPRA 37	224.0.205.37	47037	OPRA 85	224.0.205.85	47085
OPRA 38	224.0.205.38	47038	OPRA 86	224.0.205.86	47086
OPRA 39	224.0.205.39	47039	OPRA 87	224.0.205.87	47087
OPRA 40	224.0.205.40	47040	OPRA 88	224.0.205.88	47088
OPRA 41	224.0.205.41	47041	OPRA 89	224.0.205.89	47089
OPRA 42	224.0.205.42	47042	OPRA 90	224.0.205.90	47090
OPRA 43	224.0.205.43	47043	OPRA 91	224.0.205.91	47091
OPRA 44	224.0.205.44	47044	OPRA 92	224.0.205.92	47092
OPRA 45	224.0.205.45	47045	OPRA 93	224.0.205.93	47093
OPRA 46	224.0.205.46	47046	OPRA 94	224.0.205.94	47094
OPRA 47	224.0.205.47	47047	OPRA 95	224.0.205.95	47095
OPRA 48	224.0.205.48	47048	OPRA 96	224.0.205.96	47096

Appendix 2 - OPRA Regular Session Playback Test IP Multicast Feeds, (Network 'A' only) - **CONTINUED**

PRODUCT NAME:	OPRA Regular Session		
	RESERVED FOR FUTURE USE		
PLAYBACK TEST “DATA STREAM - A” SUBNET: 224.0.205.0/25			
Originated Data Line	Multicast Group Address		Multicast Group Port
OPRA 220	224.0.205.105		47105
OPRA 221	224.0.205.106		47106
OPRA 222	224.0.205.107		47107
OPRA 223	224.0.205.108		47108
OPRA 224	224.0.205.109		47109
OPRA 225	224.0.205.110		47110

Appendix 2 - OPRA Regular Session Playback Test IP Multicast Feeds, (Network 'B' only)

PRODUCT NAME:		OPRA Regular Session			
Network Subnet:		ONE (1) NETWORK SUBNET GROUP PER DATA STREAM			
PLAYBACK TEST “DATA STREAM - B” SUBNET: 224.0.207.0/25					
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 1	224.0.207.1	47001	OPRA 49	224.0.207.49	47049
OPRA 2	224.0.207.2	47002	OPRA 50	224.0.207.50	47050
OPRA 3	224.0.207.3	47003	OPRA 51	224.0.207.51	47051
OPRA 4	224.0.207.4	47004	OPRA 52	224.0.207.52	47052
OPRA 5	224.0.207.5	47005	OPRA 53	224.0.207.53	47053
OPRA 6	224.0.207.6	47006	OPRA 54	224.0.207.54	47054
OPRA 7	224.0.207.7	47007	OPRA 55	224.0.207.55	47055
OPRA 8	224.0.207.8	47008	OPRA 56	224.0.207.56	47056
OPRA 9	224.0.207.9	47009	OPRA 57	224.0.207.57	47057
OPRA 10	224.0.207.10	47010	OPRA 58	224.0.207.58	47058
OPRA 11	224.0.207.11	47011	OPRA 59	224.0.207.59	47059
OPRA 12	224.0.207.12	47012	OPRA 60	224.0.207.60	47060
OPRA 13	224.0.207.13	47013	OPRA 61	224.0.207.61	47061
OPRA 14	224.0.207.14	47014	OPRA 62	224.0.207.62	47062
OPRA 15	224.0.207.15	47015	OPRA 63	224.0.207.63	47063
OPRA 16	224.0.207.16	47016	OPRA 64	224.0.207.64	47064
OPRA 17	224.0.207.17	47017	OPRA 65	224.0.207.65	47065
OPRA 18	224.0.207.18	47018	OPRA 66	224.0.207.66	47066
OPRA 19	224.0.207.19	47019	OPRA 67	224.0.207.67	47067
OPRA 20	224.0.207.20	47020	OPRA 68	224.0.207.68	47068
OPRA 21	224.0.207.21	47021	OPRA 69	224.0.207.69	47069
OPRA 22	224.0.207.22	47022	OPRA 70	224.0.207.70	47070
OPRA 23	224.0.207.23	47023	OPRA 71	224.0.207.71	47071
OPRA 24	224.0.207.24	47024	OPRA 72	224.0.207.72	47072
OPRA 25	224.0.207.25	47025	OPRA 73	224.0.207.73	47073
OPRA 26	224.0.207.26	47026	OPRA 74	224.0.207.74	47074
OPRA 27	224.0.207.27	47027	OPRA 75	224.0.207.75	47075
OPRA 28	224.0.207.28	47028	OPRA 76	224.0.207.76	47076
OPRA 29	224.0.207.29	47029	OPRA 77	224.0.207.77	47077
OPRA 30	224.0.207.30	47030	OPRA 78	224.0.207.78	47078
OPRA 31	224.0.207.31	47031	OPRA 79	224.0.207.79	47079
OPRA 32	224.0.207.32	47032	OPRA 80	224.0.207.80	47080
OPRA 33	224.0.207.33	47033	OPRA 81	224.0.207.81	47081
OPRA 34	224.0.207.34	47034	OPRA 82	224.0.207.82	47082
OPRA 35	224.0.207.35	47035	OPRA 83	224.0.207.83	47083
OPRA 36	224.0.207.36	47036	OPRA 84	224.0.207.84	47084
OPRA 37	224.0.207.37	47037	OPRA 85	224.0.207.85	47085
OPRA 38	224.0.207.38	47038	OPRA 86	224.0.207.86	47086
OPRA 39	224.0.207.39	47039	OPRA 87	224.0.207.87	47087
OPRA 40	224.0.207.40	47040	OPRA 88	224.0.207.88	47088
OPRA 41	224.0.207.41	47041	OPRA 89	224.0.207.89	47089
OPRA 42	224.0.207.42	47042	OPRA 90	224.0.207.90	47090
OPRA 43	224.0.207.43	47043	OPRA 91	224.0.207.91	47091
OPRA 44	224.0.207.44	47044	OPRA 92	224.0.207.92	47092
OPRA 45	224.0.207.45	47045	OPRA 93	224.0.207.93	47093
OPRA 46	224.0.207.46	47046	OPRA 94	224.0.207.94	47094
OPRA 47	224.0.207.47	47047	OPRA 95	224.0.207.95	47095
OPRA 48	224.0.207.48	47048	OPRA 96	224.0.207.96	47096

Appendix 2 - OPRA Regular Session Playback Test IP Multicast Feeds, (Network 'B' only) - CONTINUED

PRODUCT NAME:	OPRA Regular Session	
	RESERVED FOR FUTURE USE	
PLAYBACK TEST “DATA STREAM - B” SUBNET: 224.0.207.0/25		
Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 220	224.0.207.105	47105
OPRA 221	224.0.207.106	47106
OPRA 222	224.0.207.107	47107
OPRA 223	224.0.207.108	47108
OPRA 224	224.0.207.109	47109
OPRA 225	224.0.207.110	47110

8 Appendix 3 - OPRA Global Trading Hours (GTH) Production, Real-Time IP Multicast Feeds, Dual Sets (Network 'A' / 'B' Data Streams)

PRODUCT NAME:	OPRA Global Trading Hours (GTH)	
NETWORK SUBNETS:	ONE (1) NETWORK SUBNET GROUP PER DATA STREAM	
PRODUCTION “DATA STREAM - A” SUBNET: 224.0.200.0/28		
Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 201	224.0.200.1	48001
OPRA 202	224.0.200.2	48002
OPRA 203	224.0.200.3	48003
OPRA 204	224.0.200.4	48004
PRODUCTION “DATA STREAM - B” SUBNET: 224.0.200.128/28		
Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 201	224.0.200.128	48001
OPRA 202	224.0.200.129	48002
OPRA 203	224.0.200.130	48003
OPRA 204	224.0.200.131	48004

Appendix 3 - OPRA Global Trading Hours (GTH) Retransmission IP Multicast Feeds, Dual Sets (Network 'A' / 'B' Data Streams)

PRODUCT NAME:	OPRA Global Trading Hours (GTH)	
NETWORK SUBNETS:	ONE (1) NETWORK SUBNET GROUP PER DATA STREAM	
RETRANSMISSION “DATA STREAM - A” SUBNET: 224.0.200.16/28		
Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 201	224.0.200.16	49001
OPRA 202	224.0.200.17	49002
OPRA 203	224.0.200.18	49003
OPRA 204	224.0.200.19	49004
RETRANSMISSION “DATA STREAM - B” SUBNET: 224.0.200.144/28		
Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 201	224.0.200.144	49001
OPRA 202	224.0.200.145	49002
OPRA 203	224.0.200.146	49003
OPRA 204	224.0.200.147	49004

Appendix 3 - OPRA Global Trading Hours (GTH) Playback Test IP Multicast Feeds, Dual Sets (Network 'A' / 'B' Data Streams)

PRODUCT NAME:	OPRA Global Trading Hours (GTH)	
NETWORK SUBNETS:	ONE (1) NETWORK SUBNET GROUP PER DATA STREAM	
PLAYBACK TEST “DATA STREAM - A” SUBNET: 224.0.200.32/28		
Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 201	224.0.200.32	50001
OPRA 202	224.0.200.33	50002
OPRA 203	224.0.200.34	50003
OPRA 204	224.0.200.35	50004
PLAYBACK TEST “DATA STREAM - B” SUBNET: 224.0.200.160/28		
Originated Data Line	Multicast Group Address	Multicast Group Port
OPRA 201	224.0.200.160	50001
OPRA 202	224.0.200.161	50002
OPRA 203	224.0.200.162	50003
OPRA 204	224.0.200.163	50004

9 Appendix 4 - CQS Production, Real-Time IP Multicast Feeds, Dual Sets (Network 'A' / Network 'B')

MULTICAST DATA: PRODUCTION "A-STREAM" & PRODUCTION "B-STREAM"

PRODUCT NAME:		CQS			
NETWORK SUBNETS:		TWO (2) NETWORK SUBNETS GROUPS PER DATA STREAM			
PRODUCTION “DATA STREAM - A” SUBNETS:					
224.0.203.0/28			224.0.203.32/28		
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CQS 1 / Tape A	224.0.203.1	45001	CQS 1 / Tape B	224.0.203.32	45001
CQS 2 / Tape A	224.0.203.2	45002	CQS 2 / Tape B	224.0.203.33	45002
CQS 3 / Tape A	224.0.203.3	45003	CQS 3 / Tape B	224.0.203.34	45003
CQS 4 / Tape A	224.0.203.4	45004	CQS 4 / Tape B	224.0.203.35	45004
CQS 5 / Tape A	224.0.203.5	45005	CQS 5 / Tape B	224.0.203.36	45005
CQS 6 / Tape A	224.0.203.6	45006	CQS 6 / Tape B	224.0.203.37	45006
CQS 7 / Tape A	224.0.203.7	45007	CQS 7 / Tape B	224.0.203.38	45007
CQS 8 / Tape A	224.0.203.8	45008	CQS 8 / Tape B	224.0.203.39	45008
CQS 9 / Tape A	224.0.203.9	45009	CQS 9 / Tape B	224.0.203.40	45009
CQS 10 / Tape A	224.0.203.10	45010	CQS 10 / Tape B	224.0.203.41	45010
CQS 11 / Tape A	224.0.203.11	45011	CQS 11 / Tape B	224.0.203.42	45011
CQS 12 / Tape A	224.0.203.12	45012	CQS 12 / Tape B	224.0.203.43	45012
PRODUCTION “DATA STREAM - B” SUBNETS:					
224.0.203.128/28			224.0.203.160/28		
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CQS 1 / Tape A	224.0.203.128	45001	CQS 1 / Tape B	224.0.203.160	45001
CQS 2 / Tape A	224.0.203.129	45002	CQS 2 / Tape B	224.0.203.161	45002
CQS 3 / Tape A	224.0.203.130	45003	CQS 3 / Tape B	224.0.203.162	45003
CQS 4 / Tape A	224.0.203.131	45004	CQS 4 / Tape B	224.0.203.163	45004
CQS 5 / Tape A	224.0.203.132	45005	CQS 5 / Tape B	224.0.203.164	45005
CQS 6 / Tape A	224.0.203.133	45006	CQS 6 / Tape B	224.0.203.165	45006
CQS 7 / Tape A	224.0.203.134	45007	CQS 7 / Tape B	224.0.203.166	45007
CQS 8 / Tape A	224.0.203.135	45008	CQS 8 / Tape B	224.0.203.167	45008
CQS 9 / Tape A	224.0.203.136	45009	CQS 9 / Tape B	224.0.203.168	45009
CQS 10 / Tape A	224.0.203.137	45010	CQS 10 / Tape B	224.0.203.169	45010
CQS 11 / Tape A	224.0.203.138	45011	CQS 11 / Tape B	224.0.203.170	45011
CQS 12 / Tape A	224.0.203.139	45012	CQS 12 / Tape B	224.0.203.171	45012

Appendix 4 - CQS Retransmission IP Multicast Feeds, Dual Set (Network 'A' / Network 'B')

MULTICAST DATA: RETRANSMISSION GROUP, DUAL SET

PRODUCT NAME:		CQS			
NETWORK SUBNETS:		TWO (2) NETWORK SUBNETS GROUPS PER DATA STREAM			
RETRANSMISSION “DATA STREAM - A” SUBNETS:					
224.0.203.64/28			224.0.203.96/28		
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CQS 1 / Tape A	224.0.203.64	46001	CQS 1 / Tape B	224.0.203.96	46001
CQS 2 / Tape A	224.0.203.65	46002	CQS 2 / Tape B	224.0.203.97	46002
CQS 3 / Tape A	224.0.203.66	46003	CQS 3 / Tape B	224.0.203.98	46003
CQS 4 / Tape A	224.0.203.67	46004	CQS 4 / Tape B	224.0.203.99	46004
CQS 5 / Tape A	224.0.203.68	46005	CQS 5 / Tape B	224.0.203.100	46005
CQS 6 / Tape A	224.0.203.69	46006	CQS 6 / Tape B	224.0.203.101	46006
CQS 7 / Tape A	224.0.203.70	46007	CQS 7 / Tape B	224.0.203.102	46007
CQS 8 / Tape A	224.0.203.71	46008	CQS 8 / Tape B	224.0.203.103	46008
CQS 9 / Tape A	224.0.203.72	46009	CQS 9 / Tape B	224.0.203.104	46009
CQS 10 / Tape A	224.0.203.73	46010	CQS 10 / Tape B	224.0.203.105	46010
CQS 11 / Tape A	224.0.203.74	46011	CQS 11 / Tape B	224.0.203.106	46011
CQS 12 / Tape A	224.0.203.75	46012	CQS 12 / Tape B	224.0.203.107	46012
RETRANSMISSION “DATA STREAM - B” SUBNETS:					
224.0.203.192/28			224.0.203.224/28		
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CQS 1 / Tape A	224.0.203.192	46001	CQS 1 / Tape B	224.0.203.224	46001
CQS 2 / Tape A	224.0.203.193	46002	CQS 2 / Tape B	224.0.203.225	46002
CQS 3 / Tape A	224.0.203.194	46003	CQS 3 / Tape B	224.0.203.226	46003
CQS 4 / Tape A	224.0.203.195	46004	CQS 4 / Tape B	224.0.203.227	46004
CQS 5 / Tape A	224.0.203.196	46005	CQS 5 / Tape B	224.0.203.228	46005
CQS 6 / Tape A	224.0.203.197	46006	CQS 6 / Tape B	224.0.203.229	46006
CQS 7 / Tape A	224.0.203.198	46007	CQS 7 / Tape B	224.0.203.230	46007
CQS 8 / Tape A	224.0.203.199	46008	CQS 8 / Tape B	224.0.203.231	46008
CQS 9 / Tape A	224.0.203.200	46009	CQS 9 / Tape B	224.0.203.232	46009
CQS 10 / Tape A	224.0.203.201	46010	CQS 10 / Tape B	224.0.203.233	46010
CQS 11 / Tape A	224.0.203.202	46011	CQS 11 / Tape B	224.0.203.234	46011
CQS 12 / Tape A	224.0.203.203	46012	CQS 12 / Tape B	224.0.203.235	46012

Appendix 4 - CQS Playback Test IP Multicast Feeds, Dual Set (Network 'A' / Network 'B')

MULTICAST DATA: PLAYBACK TEST GROUP, DUAL SET

PRODUCT NAME:		CQS			
NETWORK SUBNETS:		TWO (2) NETWORK SUBNETS GROUPS PER DATA STREAM			
PLAYBACK TEST “DATA STREAM - A” SUBNETS:					
224.0.200.60/31			224.0.200.62/31		
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CQS 1 / Tape A	224.0.200.60	47001	CQS 1 / Tape B	224.0.200.62	47001
CQS 2 / Tape A	224.0.200.60	47002	CQS 2 / Tape B	224.0.200.62	47002
CQS 3 / Tape A	224.0.200.60	47003	CQS 3 / Tape B	224.0.200.62	47003
CQS 4 / Tape A	224.0.200.60	47004	CQS 4 / Tape B	224.0.200.62	47004
CQS 5 / Tape A	224.0.200.60	47005	CQS 5 / Tape B	224.0.200.62	47005
CQS 6 / Tape A	224.0.200.60	47006	CQS 6 / Tape B	224.0.200.62	47006
CQS 7 / Tape A	224.0.200.60	47007	CQS 7 / Tape B	224.0.200.62	47007
CQS 8 / Tape A	224.0.200.60	47008	CQS 8 / Tape B	224.0.200.62	47008
CQS 9 / Tape A	224.0.200.60	47009	CQS 9 / Tape B	224.0.200.62	47009
CQS 10 / Tape A	224.0.200.60	47010	CQS 10 / Tape B	224.0.200.62	47010
CQS 11 / Tape A	224.0.200.60	47011	CQS 11 / Tape B	224.0.200.62	47011
CQS 12 / Tape A	224.0.200.60	47012	CQS 12 / Tape B	224.0.200.62	47012
PLAYBACK TEST “DATA STREAM - B” SUBNETS:					
224.0.200.188/31			224.0.200.190/31		
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CQS 1 / Tape A	224.0.200.188	47001	CQS 1 / Tape B	224.0.200.190	47001
CQS 2 / Tape A	224.0.200.188	47002	CQS 2 / Tape B	224.0.200.190	47002
CQS 3 / Tape A	224.0.200.188	47003	CQS 3 / Tape B	224.0.200.190	47003
CQS 4 / Tape A	224.0.200.188	47004	CQS 4 / Tape B	224.0.200.190	47004
CQS 5 / Tape A	224.0.200.188	47005	CQS 5 / Tape B	224.0.200.190	47005
CQS 6 / Tape A	224.0.200.188	47006	CQS 6 / Tape B	224.0.200.190	47006
CQS 7 / Tape A	224.0.200.188	47007	CQS 7 / Tape B	224.0.200.190	47007
CQS 8 / Tape A	224.0.200.188	47008	CQS 8 / Tape B	224.0.200.190	47008
CQS 9 / Tape A	224.0.200.188	47009	CQS 9 / Tape B	224.0.200.190	47009
CQS 10 / Tape A	224.0.200.188	47010	CQS 10 / Tape B	224.0.200.190	47010
CQS 11 / Tape A	224.0.200.188	47011	CQS 11 / Tape B	224.0.200.190	47011
CQS 12 / Tape A	224.0.200.188	47012	CQS 12 / Tape B	224.0.200.190	47012

Appendix 4 - CQS Snapshot Retransmission IP Multicast Feeds, Dual Set (Network 'A' / Network 'B')

MULTICAST DATA: SNAPSHOT RETRANSMISSION GROUP, DUAL SET

PRODUCT NAME:		CQS			
NETWORK SUBNETS:		TWO (2) NETWORK SUBNETS GROUPS PER DATA STREAM			
SNAPSHOT RETRANSMISSION “DATA STREAM - A” SUBNETS:					
224.0.201.0/27			224.0.201.32/27		
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CQS 1 / Tape A	224.0.201.1	48001	CQS 1 / Tape B	224.0.201.32	48001
CQS 2 / Tape A	224.0.201.2	48002	CQS 2 / Tape B	224.0.201.33	48002
CQS 3 / Tape A	224.0.201.3	48003	CQS 3 / Tape B	224.0.201.34	48003
CQS 4 / Tape A	224.0.201.4	48004	CQS 4 / Tape B	224.0.201.35	48004
CQS 5 / Tape A	224.0.201.5	48005	CQS 5 / Tape B	224.0.201.36	48005
CQS 6 / Tape A	224.0.201.6	48006	CQS 6 / Tape B	224.0.201.37	48006
CQS 7 / Tape A	224.0.201.7	48007	CQS 7 / Tape B	224.0.201.38	48007
CQS 8 / Tape A	224.0.201.8	48008	CQS 8 / Tape B	224.0.201.39	48008
CQS 9 / Tape A	224.0.201.9	48009	CQS 9 / Tape B	224.0.201.40	48009
CQS 10 / Tape A	224.0.201.10	48010	CQS 10 / Tape B	224.0.201.41	48010
CQS 11 / Tape A	224.0.201.11	48011	CQS 11 / Tape B	224.0.201.42	48011
CQS 12 / Tape A	224.0.201.12	48012	CQS 12 / Tape B	224.0.201.43	48012
SNAPSHOT RETRANSMISSION “DATA STREAM - B” SUBNETS:					
224.0.201.128/27			224.0.201.160/27		
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CQS 1 / Tape A	224.0.201.128	48001	CQS 1 / Tape B	224.0.201.160	48001
CQS 2 / Tape A	224.0.201.129	48002	CQS 2 / Tape B	224.0.201.161	48002
CQS 3 / Tape A	224.0.201.130	48003	CQS 3 / Tape B	224.0.201.162	48003
CQS 4 / Tape A	224.0.201.131	48004	CQS 4 / Tape B	224.0.201.163	48004
CQS 5 / Tape A	224.0.201.132	48005	CQS 5 / Tape B	224.0.201.164	48005
CQS 6 / Tape A	224.0.201.133	48006	CQS 6 / Tape B	224.0.201.165	48006
CQS 7 / Tape A	224.0.201.134	48007	CQS 7 / Tape B	224.0.201.166	48007
CQS 8 / Tape A	224.0.201.135	48008	CQS 8 / Tape B	224.0.201.167	48008
CQS 9 / Tape A	224.0.201.136	48009	CQS 9 / Tape B	224.0.201.168	48009
CQS 10 / Tape A	224.0.201.137	48010	CQS 10 / Tape B	224.0.201.169	48010
CQS 11 / Tape A	224.0.201.138	48011	CQS 11 / Tape B	224.0.201.170	48011
CQS 12 / Tape A	224.0.201.139	48012	CQS 12 / Tape B	224.0.201.171	48012

10 Appendix 5 - CTS Production, Real-Time IP Multicast Feeds, Dual Sets (Network 'A' / Network 'B')

MULTICAST DATA: PRODUCTION "A-STREAM" & PRODUCTION "B-STREAM"

PRODUCT NAME:		CTS			
NETWORK SUBNETS:		THREE (3) NETWORK SUBNETS GROUPS PER DATA STREAM			
PRODUCTION “DATA STREAM - A” SUBNETS:					
224.0.202.0/28			224.0.202.32/28		224.0.200.96/30
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CTS 1 / Tape A	224.0.202.1	45001	CTS 1 / Tape B	224.0.202.32	45001
CTS 2 / Tape A	224.0.202.2	45002	CTS 2 / Tape B	224.0.202.33	45002
CTS 3 / Tape A	224.0.202.3	45003	CTS 3 / Tape B	224.0.202.34	45003
CTS 4 / Tape A	224.0.202.4	45004	CTS 4 / Tape B	224.0.202.35	45004
CTS 5 / Tape A	224.0.202.5	45005	CTS 5 / Tape B	224.0.202.36	45005
CTS 6 / Tape A	224.0.202.6	45006	CTS 6 / Tape B	224.0.202.37	45006
CTS 7 / Tape A	224.0.202.7	45007	CTS 7 / Tape B	224.0.202.38	45007
CTS 8 / Tape A	224.0.202.8	45008	CTS 8 / Tape B	224.0.202.39	45008
CTS 9 / Tape A	224.0.202.9	45009	CTS 9 / Tape B	224.0.202.40	45009
CTS 10 / Tape A	224.0.202.10	45010	CTS 10 / Tape B	224.0.202.41	45010
CTS 11 / Tape A	224.0.202.11	45011	CTS 11 / Tape B	224.0.202.42	45011
CTS 12 / Tape A	224.0.202.12	45012	CTS 12 / Tape B	224.0.202.43	45012
			CTS INDEX 1 / Tape B	224.0.200.96	45001
			CTS INDEX 2 / Tape B	224.0.200.97	45002
PRODUCTION “DATA STREAM - B” SUBNETS:					
224.0.202.128/28			224.0.202.160/28		224.0.200.224/30
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CTS 1 / Tape A	224.0.202.128	45001	CTS 1 / Tape B	224.0.202.160	45001
CTS 2 / Tape A	224.0.202.129	45002	CTS 2 / Tape B	224.0.202.161	45002
CTS 3 / Tape A	224.0.202.130	45003	CTS 3 / Tape B	224.0.202.162	45003
CTS 4 / Tape A	224.0.202.131	45004	CTS 4 / Tape B	224.0.202.163	45004
CTS 5 / Tape A	224.0.202.132	45005	CTS 5 / Tape B	224.0.202.164	45005
CTS 6 / Tape A	224.0.202.133	45006	CTS 6 / Tape B	224.0.202.165	45006
CTS 7 / Tape A	224.0.202.134	45007	CTS 7 / Tape B	224.0.202.166	45007
CTS 8 / Tape A	224.0.202.135	45008	CTS 8 / Tape B	224.0.202.167	45008
CTS 9 / Tape A	224.0.202.136	45009	CTS 9 / Tape B	224.0.202.168	45009
CTS 10 / Tape A	224.0.202.137	45010	CTS 10 / Tape B	224.0.202.169	45010
CTS 11 / Tape A	224.0.202.138	45011	CTS 11 / Tape B	224.0.202.170	45011
CTS 12 / Tape A	224.0.202.139	45012	CTS 12 / Tape B	224.0.202.171	45012
			CTS INDEX 1 / Tape B	224.0.200.224	45001
			CTS INDEX 2 / Tape B	224.0.200.225	45002

Appendix 5 - CTS Retransmission IP Multicast Feeds, Dual Set (Network 'A' / Network 'B')

MULTICAST DATA: RETRANSMISSION GROUP, DUAL SET

PRODUCT NAME:		CTS			
NETWORK SUBNETS:		THREE (3) NETWORK SUBNETS GROUPS PER DATA STREAM			
RETRANSMISSION “DATA STREAM - A” SUBNETS:					
224.0.202.64/28			224.0.202.96/28		224.0.200.100/30
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CTS 1 / Tape A	224.0.202.64	46001	CTS 1 / Tape B	224.0.202.96	46001
CTS 2 / Tape A	224.0.202.65	46002	CTS 2 / Tape B	224.0.202.97	46002
CTS 3 / Tape A	224.0.202.66	46003	CTS 3 / Tape B	224.0.202.98	46003
CTS 4 / Tape A	224.0.202.67	46004	CTS 4 / Tape B	224.0.202.99	46004
CTS 5 / Tape A	224.0.202.68	46005	CTS 5 / Tape B	224.0.202.100	46005
CTS 6 / Tape A	224.0.202.69	46006	CTS 6 / Tape B	224.0.202.101	46006
CTS 7 / Tape A	224.0.202.70	46007	CTS 7 / Tape B	224.0.202.102	46007
CTS 8 / Tape A	224.0.202.71	46008	CTS 8 / Tape B	224.0.202.103	46008
CTS 9 / Tape A	224.0.202.72	46009	CTS 9 / Tape B	224.0.202.104	46009
CTS 10 / Tape A	224.0.202.73	46010	CTS 10 / Tape B	224.0.202.105	46010
CTS 11 / Tape A	224.0.202.74	46011	CTS 11 / Tape B	224.0.202.106	46011
CTS 12 / Tape A	224.0.202.75	46012	CTS 12 / Tape B	224.0.202.107	46012
			CTS INDEX 1 / Tape B	224.0.200.100	46001
			CTS INDEX 2 / Tape B	224.0.200.101	46002
RETRANSMISSION “DATA STREAM - B” SUBNETS:					
224.0.202.192/28			224.0.202.224/28		224.0.200.228/30
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CTS 1 / Tape A	224.0.202.192	46001	CTS 1 / Tape B	224.0.202.224	46001
CTS 2 / Tape A	224.0.202.193	46002	CTS 2 / Tape B	224.0.202.225	46002
CTS 3 / Tape A	224.0.202.194	46003	CTS 3 / Tape B	224.0.202.226	46003
CTS 4 / Tape A	224.0.202.195	46004	CTS 4 / Tape B	224.0.202.227	46004
CTS 5 / Tape A	224.0.202.196	46005	CTS 5 / Tape B	224.0.202.228	46005
CTS 6 / Tape A	224.0.202.197	46006	CTS 6 / Tape B	224.0.202.229	46006
CTS 7 / Tape A	224.0.202.198	46007	CTS 7 / Tape B	224.0.202.230	46007
CTS 8 / Tape A	224.0.202.199	46008	CTS 8 / Tape B	224.0.202.231	46008
CTS 9 / Tape A	224.0.202.200	46009	CTS 9 / Tape B	224.0.202.232	46009
CTS 10 / Tape A	224.0.202.201	46010	CTS 10 / Tape B	224.0.202.233	46010
CTS 11 / Tape A	224.0.202.202	46011	CTS 11 / Tape B	224.0.202.234	46011
CTS 12 / Tape A	224.0.202.203	46012	CTS 12 / Tape B	224.0.202.235	46012
			CTS INDEX 1 / Tape B	224.0.200.228	46001
			CTS INDEX 2 / Tape B	224.0.200.229	46002

Appendix 5 - CTS Playback Test IP Multicast Feeds, Dual Sets (Network 'A' / Network 'B')

MULTICAST DATA: PLAYBACK TEST GROUP, DUAL SET

PRODUCT NAME:		CTS			
NETWORK SUBNETS:		THREE (3) NETWORK SUBNETS GROUPS PER DATA STREAM			
PLAYBACK TEST “DATA STREAM - A” SUBNETS:					
224.0.200.56/31			224.0.200.58/31		224.0.200.104/30
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CTS 1 / Tape A	224.0.200.56	47001	CTS 1 / Tape B	224.0.200.58	47001
CTS 2 / Tape A	224.0.200.56	47002	CTS 2 / Tape B	224.0.200.58	47002
CTS 3 / Tape A	224.0.200.56	47003	CTS 3 / Tape B	224.0.200.58	47003
CTS 4 / Tape A	224.0.200.56	47004	CTS 4 / Tape B	224.0.200.58	47004
CTS 5 / Tape A	224.0.200.56	47005	CTS 5 / Tape B	224.0.200.58	47005
CTS 6 / Tape A	224.0.200.56	47006	CTS 6 / Tape B	224.0.200.58	47006
CTS 7 / Tape A	224.0.200.56	47007	CTS 7 / Tape B	224.0.200.58	47007
CTS 8 / Tape A	224.0.200.56	47008	CTS 8 / Tape B	224.0.200.58	47008
CTS 9 / Tape A	224.0.200.56	47009	CTS 9 / Tape B	224.0.200.58	47009
CTS 10 / Tape A	224.0.200.56	47010	CTS 10 / Tape B	224.0.200.58	47010
CTS 11 / Tape A	224.0.200.56	47011	CTS 11 / Tape B	224.0.200.58	47011
CTS 12 / Tape A	224.0.200.56	47012	CTS 12 / Tape B	224.0.200.58	47012
			CTS INDEX 1 / Tape B	224.0.200.104	47001
			CTS INDEX 2 / Tape B	224.0.200.105	47002
PLAYBACK TEST “DATA STREAM - B” SUBNETS:					
224.0.200.184/31			224.0.200.186/31		224.0.200.232/30
Originated Data Line	Multicast Group Address	Multicast Group Port	Originated Data Line	Multicast Group Address	Multicast Group Port
CTS 1 / Tape A	224.0.200.184	47001	CTS 1 / Tape B	224.0.200.186	47001
CTS 2 / Tape A	224.0.200.184	47002	CTS 2 / Tape B	224.0.200.186	47002
CTS 3 / Tape A	224.0.200.184	47003	CTS 3 / Tape B	224.0.200.186	47003
CTS 4 / Tape A	224.0.200.184	47004	CTS 4 / Tape B	224.0.200.186	47004
CTS 5 / Tape A	224.0.200.184	47005	CTS 5 / Tape B	224.0.200.186	47005
CTS 6 / Tape A	224.0.200.184	47006	CTS 6 / Tape B	224.0.200.186	47006
CTS 7 / Tape A	224.0.200.184	47007	CTS 7 / Tape B	224.0.200.186	47007
CTS 8 / Tape A	224.0.200.184	47008	CTS 8 / Tape B	224.0.200.186	47008
CTS 9 / Tape A	224.0.200.184	47009	CTS 9 / Tape B	224.0.200.186	47009
CTS 10 / Tape A	224.0.200.184	47010	CTS 10 / Tape B	224.0.200.186	47010
CTS 11 / Tape A	224.0.200.184	47011	CTS 11 / Tape B	224.0.200.186	47011
CTS 12 / Tape A	224.0.200.184	47012	CTS 12 / Tape B	224.0.200.186	47012
			CTS INDEX 1 / Tape B	224.0.200.232	47001
			CTS INDEX 2 / Tape B	224.0.200.233	47002